Global Cancer Patients Analysis (2015–2024)

Project Overview

This project analyzes a global cancer patient dataset containing more than 50,000 patient records collected between 2015 and 2024. The objective is to understand the impact of demographic characteristics, lifestyle risk factors, cancer stages, and treatment costs on cancer severity and patient survival outcomes.

The project combines Exploratory Data Analysis (EDA), Statistical Analysis, and Machine Learning techniques to uncover meaningful healthcare insights and identify key predictors of cancer severity.

Business Problem

Cancer is one of the leading causes of mortality worldwide. Healthcare organizations require data-driven insights to:

- Identify major cancer risk factors.
- Improve early-stage diagnosis rates.
- Understand treatment cost patterns.
- Predict disease severity.
- Support healthcare planning and resource allocation.

This project aims to answer these questions using historical patient data.

Dataset Information

Dataset Source

Global Cancer Patients Dataset (2015–2024)

Total Records

50,000+ Patients

Key Features

- Patient ID
- Age
- Gender
- Country/Region
- Cancer Type
- Cancer Stage
- Genetic Risk
- Air Pollution Exposure
- Alcohol Use

- Smoking Level
- Obesity Level
- Treatment Cost (USD)
- Survival Years
- Target Severity Score

Tools & Technologies Used

- Python
- Pandas
- NumPy
- Matplotlib
- Seaborn
- SciPy
- Scikit-Learn
- Google Colab

Methodology

1. Data Understanding

Initial exploration was performed using:

- Dataset Information
- Descriptive Statistics
- Missing Value Inspection

2. Exploratory Data Analysis

The following analyses were conducted:

- Age Distribution Analysis
- Gender Distribution Analysis
- Country-wise Patient Distribution
- Cancer Type Analysis
- Cancer Stage Analysis
- Treatment Cost Distribution
- Survival Year Distribution

3. Risk Factor Analysis

The impact of the following variables on cancer severity was studied:

- Genetic Risk
- Air Pollution
- Alcohol Use
- Smoking
- Obesity Level

Linear regression analysis was applied to evaluate the relationship between each risk factor and the severity score.

4. Statistical Analysis

Correlation analysis was performed using:

- Pearson Correlation
- Spearman Correlation

to identify relationships between patient characteristics and outcome variables.

5. Machine Learning

Random Forest Regression was used to:

- Predict Cancer Severity Score
- Identify the most influential predictive factors

Feature importance analysis was performed to determine which variables contribute most to disease severity.

Key Findings

Demographic Analysis

- Patient ages range from 20 to 89 years.
- Average patient age is approximately 54 years.
- The dataset contains Male, Female, and Other gender categories.
- Patient records are distributed across 10 countries, enabling global comparisons.

Cancer Distribution

- Multiple cancer types are represented, including:
- Lung Cancer
- Breast Cancer
- Prostate Cancer
- Leukemia
- Liver Cancer
- Skin Cancer

- Cervical Cancer

- Cancer Stage II was observed as the most common stage.

Risk Factor Impact

The analysis revealed positive relationships between severity score and:

- Smoking

- Genetic Risk
- Air Pollution
- Alcohol Consumption
- Obesity

Patients with higher values of these risk factors generally exhibited higher severity scores.

Early Diagnosis Analysis

Early-stage diagnosis rates (Stage 0 and Stage I) were calculated across different cancer types.

Key observation:

- Liver Cancer showed the highest early-diagnosis proportion.
- Lung Cancer showed the lowest early-diagnosis proportion.

This suggests that lung cancer screening programs may require further improvement.

Correlation Analysis

The strongest positive correlations with Cancer Severity Score were:

1. Smoking
2. Genetic Risk
3. Air Pollution
4. Alcohol Use

These findings indicate that lifestyle and environmental factors play a significant role in disease severity.

Treatment Cost Analysis

Analysis across countries, age groups, and genders showed:

- Developed countries tend to have significantly higher treatment costs.
- Gender-based cost differences are relatively small.
- Treatment cost patterns vary by country and age group.

Survival Analysis

Correlation analysis indicated that most variables showed weak direct relationships with survival years, suggesting survival outcomes may depend on more complex interactions between medical and environmental factors.

Machine Learning Results

Model Used

Random Forest Regressor

Objective

Predict Cancer Severity Score

Important Predictors Identified

1. Smoking
2. Genetic Risk
3. Treatment Cost
4. Alcohol Use
5. Air Pollution

Smoking emerged as the strongest predictor of cancer severity.

Business Insights

Healthcare Providers

- Focus on smoking cessation programs.
- Strengthen early detection initiatives, especially for lung cancer.
- Develop targeted interventions for high-risk populations.

Policy Makers

- Reduce environmental pollution exposure.
- Increase awareness campaigns regarding smoking and alcohol use.
- Expand cancer screening programs.

Researchers

- Build predictive healthcare models using identified risk factors.
- Explore advanced survival prediction techniques.

Conclusion

This project successfully demonstrates the use of data analytics and machine learning in healthcare. Through exploratory analysis, statistical testing, and predictive modeling, several important factors influencing cancer severity were identified.

The findings indicate that smoking, genetic predisposition, environmental exposure, and lifestyle habits are major contributors to cancer severity. These insights can support healthcare organizations and policymakers in developing preventive strategies and improving patient outcomes.

The project highlights practical applications of Python-based data analysis and demonstrates strong skills in EDA, statistical analysis, visualization, and machine learning.